

$$\begin{array}{ccccc}
T^2[X, Y] & \xrightarrow{\mu[X, Y]} & T[X, Y] & \xrightarrow{\xi_{X, Y}} & [TX, TY] \\
T\xi_{X, Y} \downarrow & & & & \downarrow [\mu_X, TY] \\
T[TX, TY] & \xrightarrow{\xi_{TX, TY}} & [T^2X, T^2Y] & \xrightarrow{[T^2X, \mu_Y]} & [T^2X, TY]
\end{array}$$